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23CSE463 - Quantum Computing
Capstone Problem Statements

Quantum Computing & Simulation

1. Design a basic QRNG (Quantum Random Number Generator) using a quantum circuit that utilizes superposition and measurement to generate random bits.
2. Develop a basic simulator for Quantum Key Distribution (BB84 Protocol) to demonstrate how secure communication can be achieved using quantum principles.
3. Develop a tool/app that accepts a multi-qubit quantum circuit, isolates single-qubit reduced density matrices using partial tracing, and visualizes each qubit's mixed state on the Bloch sphere.
4. Simulate a superdense coding protocol where two classical bits are transmitted using a single qubit with prior entanglement.
5. Use quantum simulation (e.g., VQE) to estimate the ground state energy of a small molecule such as Hydrogen (H_2) or LiH.
6. Build a dashboard that shows live/public quantum computing jobs from IBM Quantum.
7. Develop a Quantum System Simulation Platform to analyze multi-qubit gate operations, measurement processes, and dominant decoherence mechanisms.
8. Develop an application-driven implementation of a foundational quantum algorithm, such as Grover's Search, Quantum Amplitude Estimation, Quantum Fourier Transform, or another core quantum algorithm, and evaluate its performance, resource requirements, and practical advantages compared to a classical solution.
9. Develop an Interactive Quantum Education and Visualization Tool that helps users understand fundamental quantum computing concepts, algorithms, and phenomena through engaging visualizations, simulations, or gamified learning experiences.
10. Develop a Quantum Simulation-based Catalyst Design System to discover efficient catalyst materials.
11. Design a Quantum Simulation-based Material Discovery Platform for next-generation batteries.
12. Develop a quantum correlation simulation platform using Qiskit to study entanglement, noise effects, and quantum illumination performance compared to classical detection methods.
13. Develop a quantum chemistry simulation framework that uses quantum computing techniques to estimate molecular ground-state energies and study catalytic systems relevant to nitrogen fixation and sustainable fertilizer production.

Quantum Finance & Machine Learning Applications

14. Maximize returns while minimizing risk (Sharpe ratio) using QAOA (Quantum Approximate Optimization Algorithm).
15. Classify future market direction using a Variational Quantum Classifier (VQC).
16. Identify anomalies in transaction data using Quantum Machine Learning (QML).
17. Perform quantum path planning for delivery vehicles.
18. Develop a Quantum Machine Learning-based Credit Risk Assessment System for evaluating borrower creditworthiness.
19. Design a Quantum AI-based Fraud Detection System to identify suspicious financial transactions in real time.
20. Design a Quantum AI-based Stock Market Prediction System to forecast stock price trends.

Quantum Applications in Healthcare & Sustainability

21. Develop a Quantum Computing-based DNA Sequence Analysis System to detect mutations and hidden genomic patterns.
22. Develop a Quantum Computing-based Drug Screening System to identify promising drug candidates.
23. Develop a Quantum or Hybrid Quantum-Classical Solution that addresses a significant sustainability or social-impact challenge in areas such as energy, climate change, healthcare, agriculture, transportation, resource management, or public welfare, and demonstrate how quantum technologies can contribute to solving the problem more effectively.
24. Develop a Quantum Optimization-based Agricultural Supply Chain Management System to optimize farm-to-market logistics.
25. Develop a Quantum Vision-based Defect Detection System for automated quality inspection in manufacturing.
26. Develop a Quantum AI-based Electricity Demand Forecasting System to predict future power consumption.
27. Develop a Quantum AI-based Medical Imaging System for disease diagnosis from medical scans.
28. Design a Quantum Machine Learning-based System to predict patient health risks from clinical data.
29. Design a Quantum AI-based Crop Yield Prediction System using weather, soil, and farming data.
30. Design a Quantum Intelligence-based Agricultural Advisory System that provides weather-driven crop recommendations.